Farmer Feedback in Tricot

Redesigning the **Tricot feedback process** & field validation 2024 - 2025

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The Feedback Gap

Problem Statement

The current feedback form has not been used systematically by Tricot trial implementers.

This is a problem because **without it, TRICOT processes do not treat participants as partners** on eye level in an equitable relationship and thus are hurting the promise of TRICOT as a participatory approach to variety testing.



Team

IITA Gender & Social Inclusion Team

Leads TRICOT activities in Nigeria—organising trials, collecting data, and delivering farmer feedback—while running parallel research on community participation and social-equity impacts.



Dr. Béla Teeken
Gender and Social Inclusion team leader



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Senior Researcher



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Steering



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User Experience



Dr. Charlotte Shumann
User Researcher
Trial Designer



Juan M. Londoño
UX Designer
Trial Implementer

NRCRI Observers

Attended several sessions to shadow the user-research process; scaling the methodology to other TRICOT groups in eastern Nigeria



Dr. Tessy Madu
NRCRI National Director



Dr. Samuel Onwuka
NRCRI Researcher



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NRCRI Researcher

Methodology Overview

Human-Centered Design Roadmap

1. Scope

- Established roles & responsibilities
- Aligned on a clear goal
- Reviewed existing insights/prior research

2. Explore

- Desk research
- Design review
- User needs assessment - interviews with Tricot implementers

3. Create

- Feedback & Co- Design sessions with colleagues and Tricot implementers
- Refined prototypes

4. Validate

- User research plan
- Refined prototypes
- Fieldwork
- Fieldwork conclusions

5. Implement

- Workshop guidelines
- Integration with ClimMob
- Implementation in the field

Problem Statement → Users needs → Prototype → Fieldwork → Integration

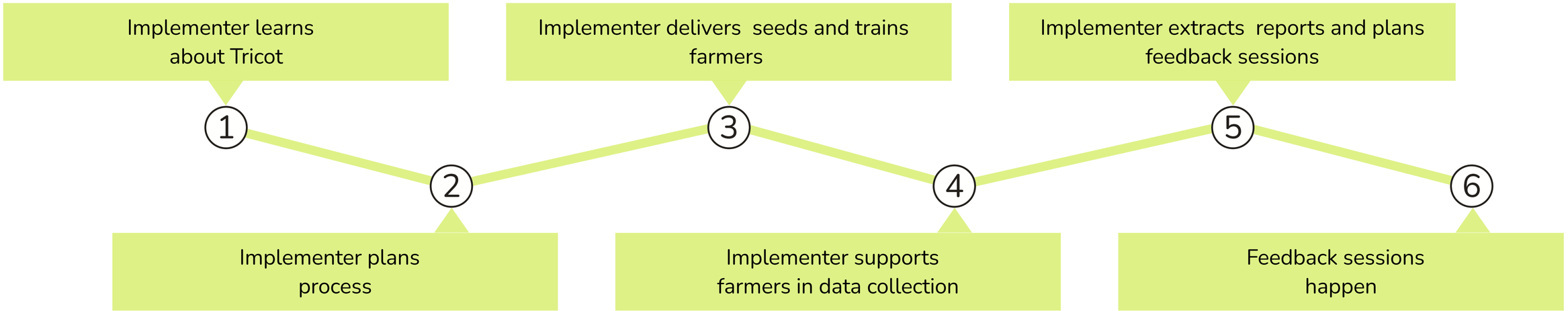
The problem does not start in the feedback sessions!

Implementers' journey with Tricot

- Implementers are not aware of feedback being an integral part of the process

- Farmers do not understand their role / don't request feedback

- No guidance on how to run feedback sessions
- Forms from Tricot are not useful and difficult to extract



- Feedback sessions are not budgeted for

- Other duties in the trial divert attention

- The form is difficult to understand and does not feel like a worthy return for farmers' participation

How can we address user needs effectively, while making sure the tricot process runs smooth and produces good data?

Co- Design sessions

Design Requirements

For farmers



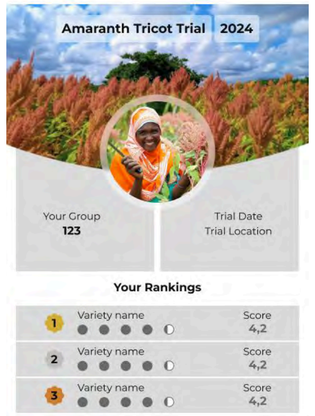
- Provide info on ranking of own varieties and offer info on traits & planting information
- Discuss next steps for farmers & address availability issue
- Use icons and visualizations to transport meaning
- Generate free discussion/reflection & interaction
- Communicate in local languages

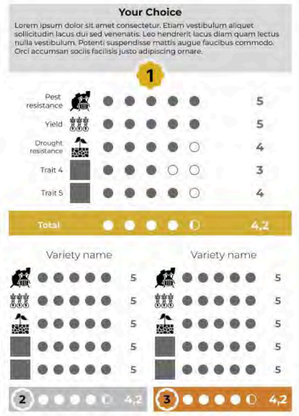
For Implementers



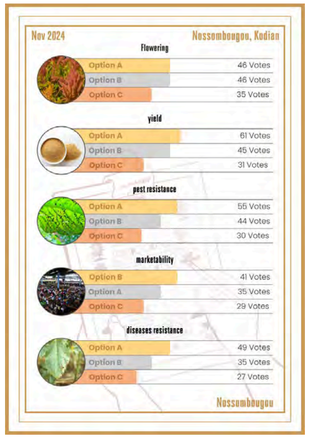
- Production needs to be cheap and easy for implementers
- Should be easily extracted from ClimMob, with ClimMob data
- Rather short meeting time for farmers

Feedback sheet ideas



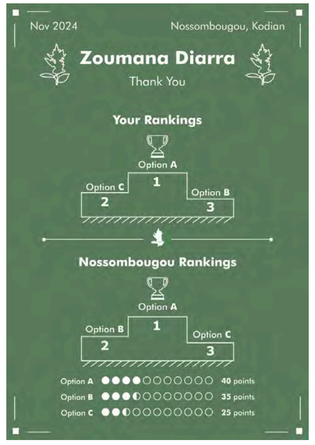




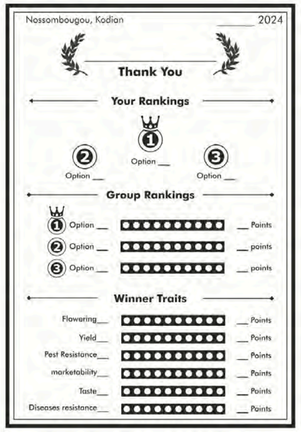














Designs that answer the key farmer and implementer requirements

Feedback-Sheet Design

For testing

Content Modules

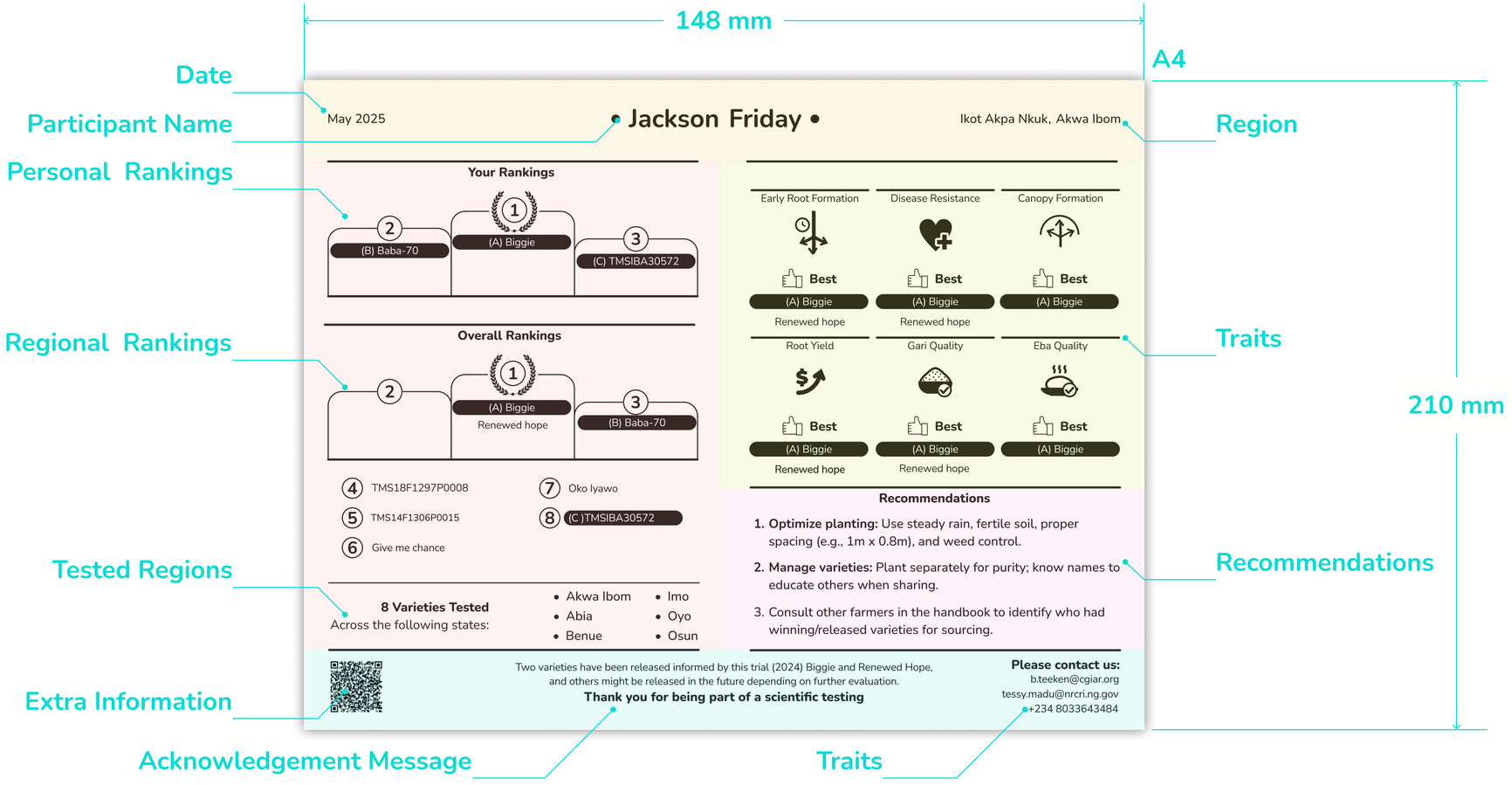
Header
- Farmer's Personal Information

Rankings Section
- Farmer's Variety Preferences
- Overall Regional Preferences

Traits Ranking
- Farmer's Traits Preferences

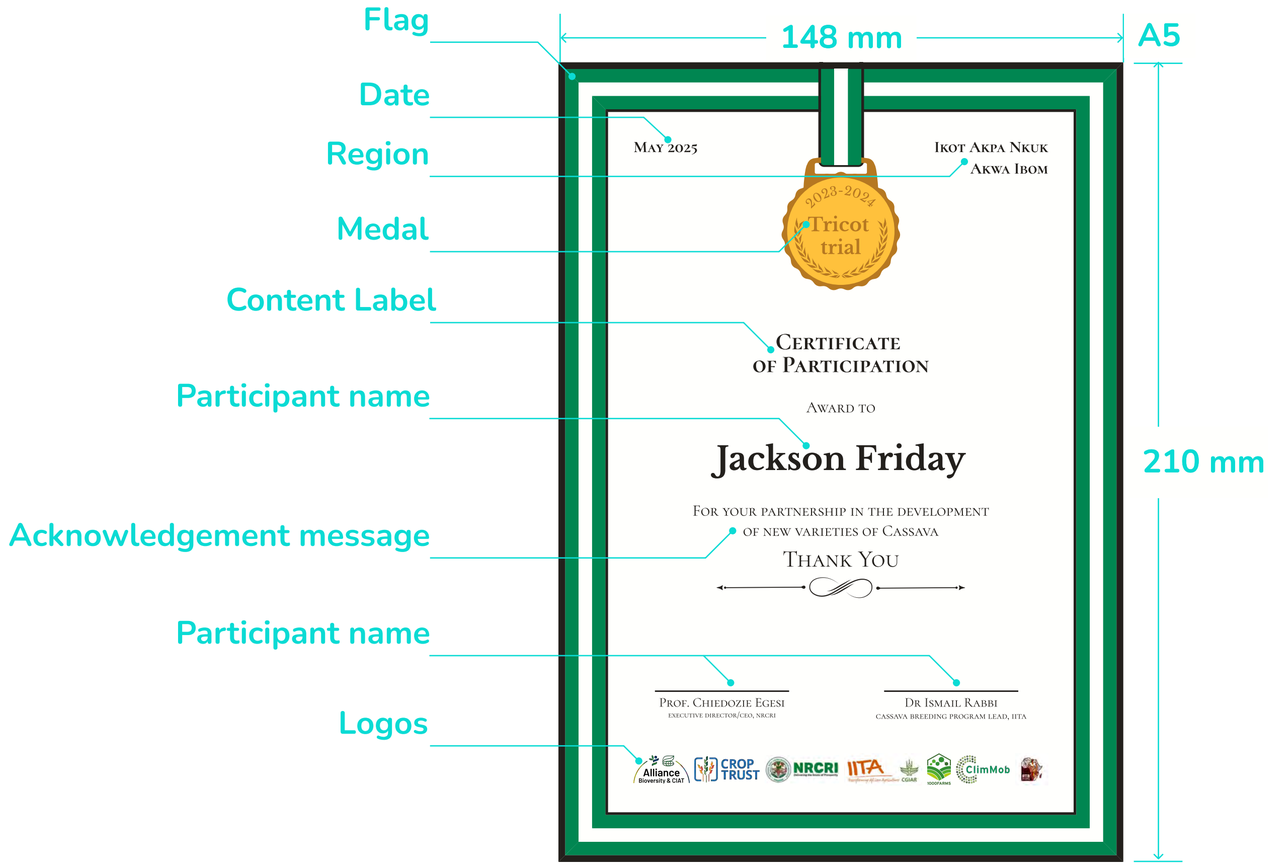
Recommendations Box
- Recommendations from Implementers.

Footer
- Communication Streams Between the Institution and the Farmers.



Diploma Design

For testing



Fieldwork

Fieldwork

The research team conducted field visits to multiple communities in Nigeria, working closely with farmers, implementers, and stakeholders.



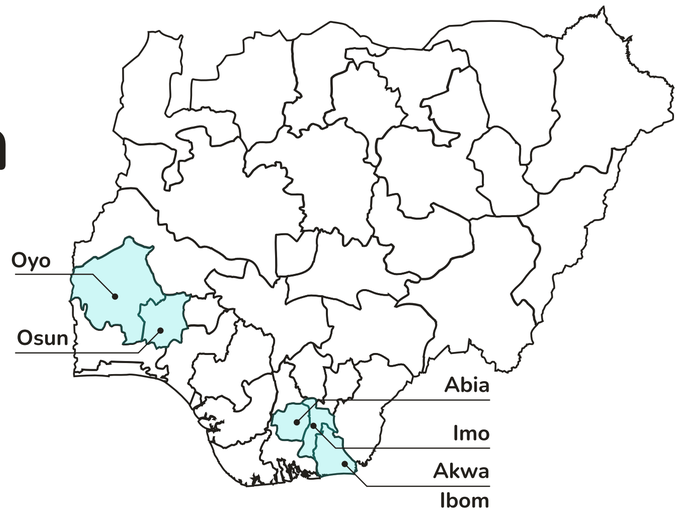






44 participants	10 Communities	5 States	3 languages
5	Agunrege	Oyo	Yoruba
5	Owotoro		-English
10	Okuku	Osun	
10	Ifon-orolu		
2	Ndoro Ibere	Abia	Igbo
2	Umuosu		-English
3	Umualumaku	Imo	
2	Ubomiri		
2	Ikoy Akai	Akwa Ibom	
3	Ikot Enang		

Nigeria



Research questions, and metrics.

	Research Question	Metrics
1.	How will the new sheet design make feedback sessions more attractive for Tricot implementers?	Qualitative info, interview analysis
2.	Does the diploma convey a sense of gratification and acknowledgement to the users?	Qualitative info, observation
3.	Do farmers understand the visualizations on the new sheet? Are farmers interested in the information on the new sheet?	Answers success rate: X% Ease of use/ interest scale

	Research Question	Metrics
4.	How do farmers come to the sessions and what do they expect?	Qualitative info, interpretation of User Journeys
5.	What did you learn about the usability and usefulness of the sheet?	Qualitative info, analysis of the next steps accounts
6.	Benchmarking with standardized UX indicators (NPS score)	Farmers' NPS Implementers NPS
7.	Do farmers understand the messages to be delivered by the icons?	Comprehension scale Matching rate

1. How will the new sheet design make feedback sessions more attractive for Tricot implementers?









In the Tricot context, the feedback sheet plus diploma act as a communication token: one page that visually summarizes each farmer's results, validates their contribution, facilitating communication with the implementer.

- **Effective sessions**

  Summarizing information and providing tangible elements enhances communication with farmers. Furthermore, using a single sheet of paper simplifies transport and reduces costs.

  "Well-designed sheet lets me talk less because farmers grasp information visually."

  "One sheet saves time ... created more time for discussion ... anything that makes work easier, I want to adopt it."

  "The the whole process, you know, has been simplified and then it's cost effective."

  "the lead farmers could also... use those feedback sheets to communicate these results."

- **Evidence for authorities**

  Clear evidence of farmers' preferences for the variety release committee.

  "serves as documented proof that farmers are truly at the center,"

  "successfully condenses 12-month data," making it easier to present to the committee.

- **Pain Points**

  Use and usability The feedback sheet's icons and symbols should be redesigned for better contextual adjustment.

  "Currency symbol should match each country"

  "Icons didn't resonate... sheet couldn't stand alone"

  "Without any explanation... looked complex"

  logistics & Timing Elements such as printing, data curation, signatures, budgeting inclusion can delay workshop arrangements.

  "Still waiting weeks later for signatures"

  "Printing can be handled... budget line should be built in"

2. Does the diploma convey a sense of gratification and acknowledgement to the users?









Yes

Having a tangible certificate made them feel that their effort was officially acknowledged—both by their peers and by the implementing team

Observations

Farmers' Reactions

- Mix understanding of the symbols.
- Generalized smile.
- Appreciation, clapping, overall good mood.

"The certificate stands as a confidence booster and proof of being able to teach other farmers".

Farmers' Expectations

- Farmers initially did not anticipate receiving a diploma.
- Upon receiving it, they immediately saw its value and began r**equesting larger, framed versions suitable for display.**

Implementers Behaviour

- The implementers decided to create a brief discovery act before explaining the diploma.
- They manifested their approval for the diploma as symbol of appreciation.
- **The breeders** in charge of the signatures decided to **sign the diploma by hand**, because they recognize it as a **symbol of appreciation**.

Challenges

- Diploma printing costs must be budgeted into the project
- Signature collection process can complicate and delay delivery
- Automation must accommodate changing logos, flags, and branding elements

3a. Are farmers interested in the information on the new sheet?

Before any previous explanations.

Yes

34 of the 44 farmers (≈ 70 %) marked the top two points on the scale (4 or 5).

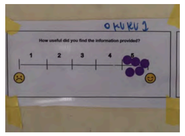
No one chose the lowest option (1), and only two farmers (≈5 %) gave a “2,” suggesting very few found the sheet information not useful.

Methodology

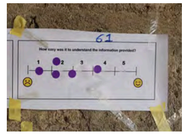
Farmers were asked to indicate their understanding/usefulness of the information on the sheet by placing a dot on a scale that was displayed on the wall.

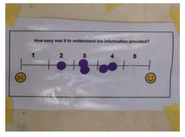
How useful did you find the information provided?





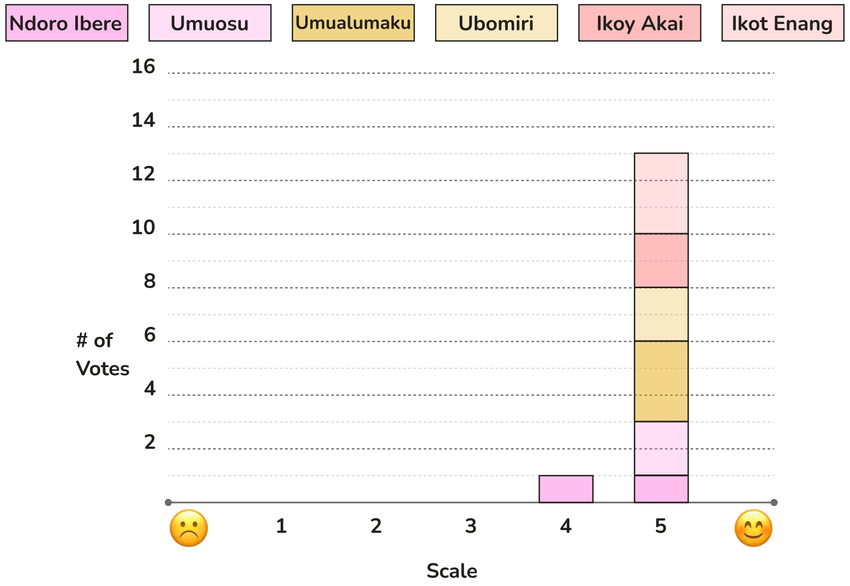
How easy was it to understand the information provided?

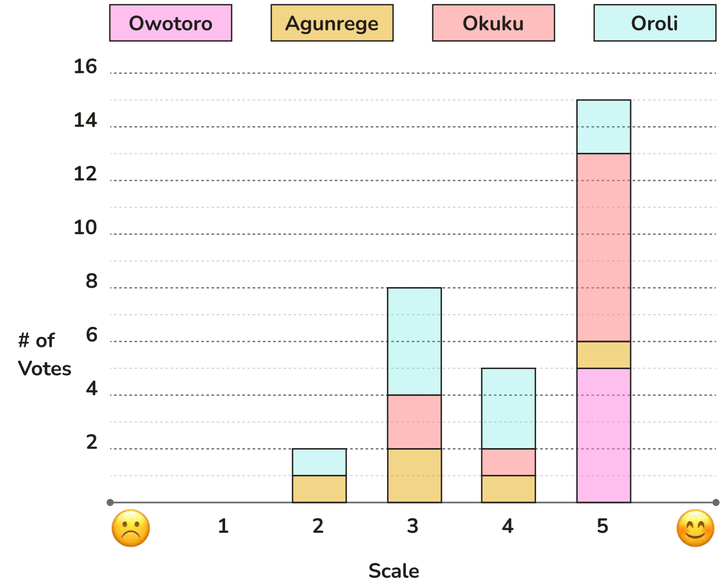




Usefulness scale

How useful did you find the information provided?





3b. Do farmers understand the visualizations on the new sheet?

Before any previous explanations.

Partially

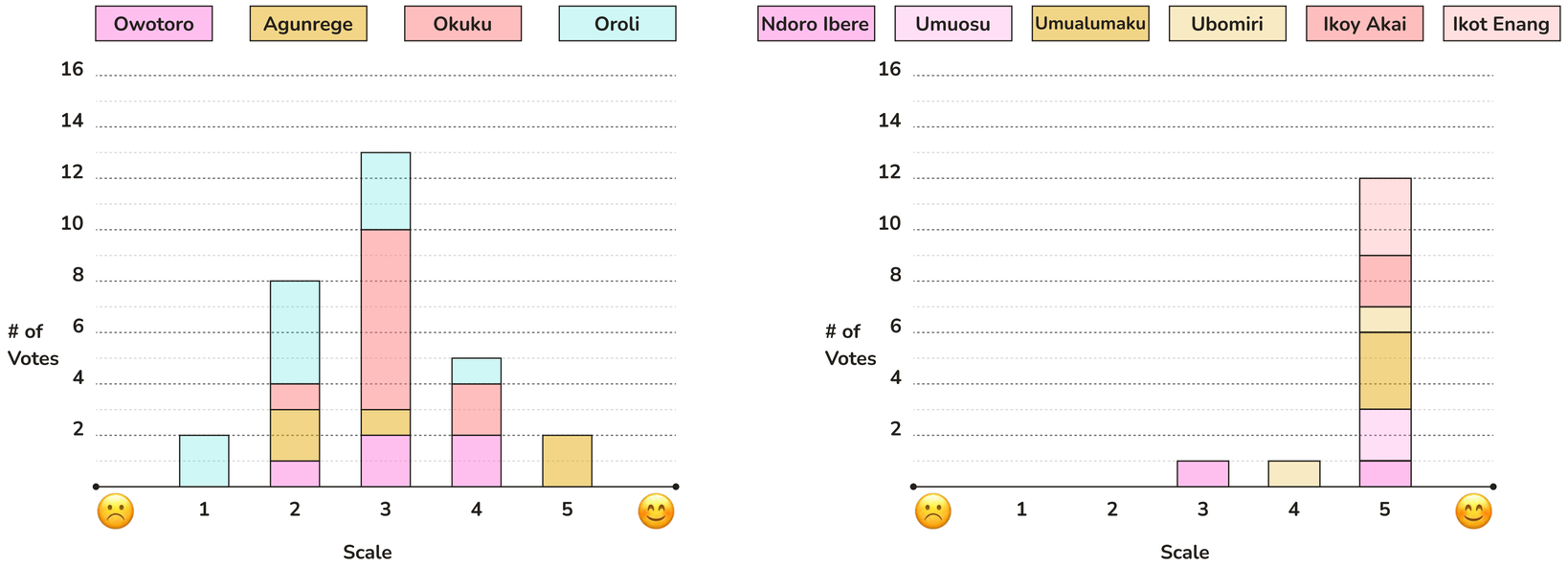
14 farmers (≈ 32 %) placed themselves in the middle. The median is exactly 3, showing that most farmers still found it only “moderately” easy to understand.

10 of 44 respondents (≈ 23 %) chose 1 or 2, signalling real difficulty decoding parts of the sheet.

20 farmers (≈ 45 %) scored 4 or 5; (literacy, prior exposure, facilitation style)

Understanding scale

How easy was it to understand the information provided?



Observations

- **Farmers’ reactions**

some farmers never seemed to engage deeply with the feedback sheet analysis. They, skimmed the page quickly and retake it after being asked by the implementer. while the implementer was distracted they urge to ask for explanation to their peers.

- **Implementers comments:**

Some farmers showed reticent to answer questions directly, gave, or aks their peers before analysis.

Challenges

- Feedback sheet was physically damaged after workshop because the constant manipulation.
- LLM aren’t accurate enough yet to adequately translate local languages like Igbo or Yoruba.
- Feedback sheet costs must be acknowledge at the beginning ot the tricot budgeting.
- Font size, and accessibility elements must be heavily revised.

4. How do farmers come to the sessions and what do they expect?

Overall sentiment toward the Tricot experience is **positive**

Selection





Happy/skeptical

- Farmers may either volunteer or be selected for participation in trials.
- Both the land they own and their farming experience are considered during the selection process.
- Participation implies a close relationship with a respected institution (IITA/NRCRI).
- Farmers are concern regarding the trial expenses in terms of land use and supplies.
- Farmers set aside personal/work/ school time to wait for the team.

Planting





Happy/expectant

- Farmers were instructed to adjust the planting space, which sparked curiosity about the reasoning behind the new method.
- Planting is a collaborative activity among the farmers.
- Planting is an expectation stage; worries about stem performance.
- First contact with stems: farmers evaluate physical characteristics (shapes expectations).

Data Collection





Happy/hopeful

- Farmers tend to respond to calls for information and share data during various growing stages, including after harvest.
- The level of skepticism and excitement about the trial varies depending on the crop's performance.
- Questionnaires used during these trials tend to be repetitive.
- Farmers establish a connection between providing quality responses and the performance of future good crop varieties.

Harvest



Proud

- High yields are associated with a sense of pride among the farmers.
- Harvesting is a collaborative effort.
- Increased yields boost motivation while helping to overcome initial skepticism.
- Harvest is a moment of revelation.
- Good harvest draws attention from other community actors.

Food Quality



Delighted/Loving

- Gari/Eba quality (colour, texture, taste) is the ultimate success metric, directly affecting household income from sales.
- Food processing is a collaborative effort.
- Unprocessed cassava (gari, eba) is rarely consumed.

Insights

- Their judgement of the programme is tightly linked to agronomic performance.
- High yields in Tricot trigger a series of new demands related to crop scalability like stem multiplication and investment.
- Planting, data collection and food-quality tests are activities done in groups. Acknowledging the single participant overlooks the team effort.

5. What did you learn about the usability and usefulness of the sheet?

The sheet acts as a **memory trigger**, **facilitating farmer-led discussions, lightening the implementer's workload,** and reinforcing the sheet's role as a Communication Token.

- Future redesigns should focus on material durability, ensure women's inclusion, and outline clear next steps.
- The workshop should be scheduled immediately after the trial to retain fresh memories.

Methodology

After explaining the sheet...

Farmers groups
Farmers formed groups based on their own variety.

Task
Implementers promote the discussion based around the following questions:

How are you going to use the information received today?

What are your personal next steps?

Observations





Self-Sorting
After the overall introduction by the implementers, farmers grouped themselves according to their top-ranked variety. Little to no assistance was needed.

Memory Trigger
Farmers shared detailed memories of the trial, leading to lively conversations and discussions. They took a defensive stance regarding their choices, articulating why they believed their selected variety was the best. There was no confrontation; instead, each group complemented the other's knowledge about different varieties.

Ownership
Once the information sheets were handed over, they were treated as personal property. The implementers no longer needed to explain the individual data line by line.

Challenges

Past Results vs. Future Plans
Despite a prompt to discuss "next steps," the conversation tended to focus on past experiences, such as yield performance, stem scarcity, and harvest issues. This shift may have been due to the **lengthy period between the trial and the feedback session**, during which the "next steps" were already implemented.

Uneven Participation
In some groups a single participant dominated the talk. Participation was gender-skewed (men spoke more). Implementers had to actively invite quieter farmers, often women to share.

Physical Durability

Sheets were constantly folded for pocketing; by session end many looked damaged or wrinkled.





6. Will the new sheet design make feedback sessions more attractive for tricot implementers/farmers?

Partially

- Farmers 70.45 NPS show clear evidence that the **design resonates and adds value.**
- Implementers with a 33.3 NPS mostly contain passives and no detractors, indicating that **small refinements are necessary.**

Methodology

Net Promoter Score (NPS)
NPS is a measure of customer loyalty and preference. It asks users:

how likely they are to recommend a product or service on a scale from 0 to 10.

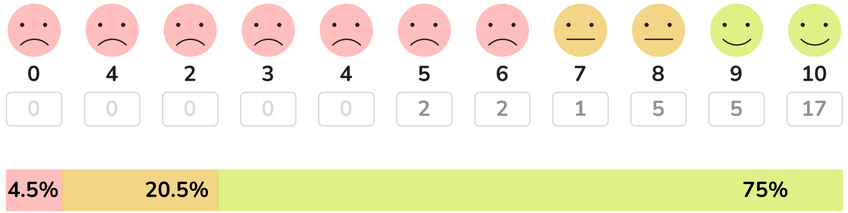
The score
NPS categorizes participants into three groups:

Detractors: These are dissatisfied users who are unlikely to recommend the use of the feedback sheet.

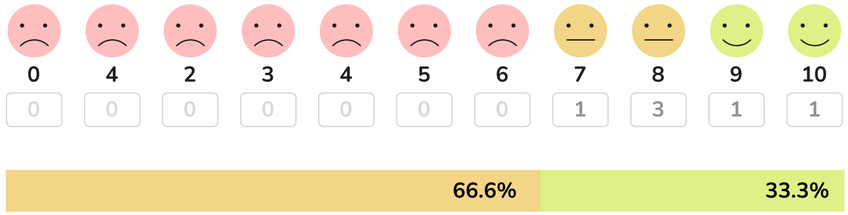
Passives: These users are neutral; they may be willing to recommend the feedback sheet but are not enthusiastic about it.

Promoters: These satisfied users are likely to recommend the use of the feedback sheet.

Farmers' NPS Score: **70.45 (Excellent)**



Implementers' NPS Score: **33.3 (Good)**



What's driving the scores - observations

Farmers

By the time the farmers filled out the scale, they had explored the sheet, engaged in discussions in peer groups, and received clarifications. As a result, most of their doubts were largely resolved.

Tangible Takeaway

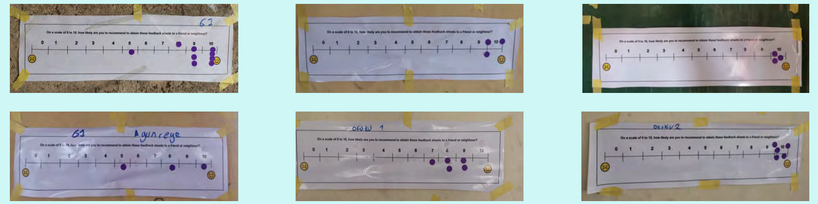
At this stage, the farmers had something to take home and share with their families and neighbors, reinforcing their sense of pride and perceived value.

Implementers

Combining user research tasks with the standard feedback meeting extended the agenda and blurred the objectives, leading to fatigue and some skepticism.

Design objections remained strong

Repeated comments about small fonts, fragile paper, and a few unclear icons kept several staff members in a passive zone.

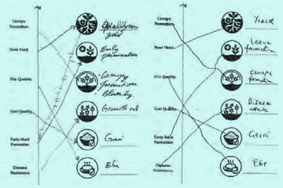


7. Do farmers understand the messages to be delivered by the icons?

Partially

- Elaborate pictograms perform better than simpler ones in every aspect.
- Icons for visible, everyday traits are clear (>4 / 5), but below-ground or abstract traits score ≤2 / 5 - Indicating redesign plus supportive text.
- Combining visuals with labels helps bridge literacy gaps and avoids relying solely on pictograms.

Methodology



12

Twelve farmers were asked to describe an icon using one or two words.

1 or 2

Farmers were asked to evaluate one of the two icon styles.

Step 1

Free association:
each icon was described in one or two words.

Step 2

Matching:
the same icons were then linked to a unordered list of trait labels.

Redesign 0 -3		Refine 3 - 4		Keep 4-5	
Root Yield	Disease Resistance	Early Root Formation	Eba Quality	Canopy Formation	Gari Quality
					
1.3	1.6	2.4	4.4	4.4	4.9
					
1.9	2.8	1.5	4.1	4.4	4.6

Pictogram exact match success rate		Line connection success rate	
	Figurative-> 50%		Figurative-> 71.5%
	Abstract-> 47.2%		Abstract-> 75.0%

Observations



- Symbols such as dollar signs, crosses, hearts, and clocks did not convey the expected meanings.
- The interpretation of symbols is highly dependent on context. For example, income from cassava comes from processed materials like gari or eba, which possess distinct characteristics such as brightness or texture. Therefore, a higher yield does not necessarily equate to increased income.
- Since cassava is a root vegetable, the presence of cassava leaves or fields does not inherently relate to yield or disease outcomes.

Feedback Sheet Conclusions

Redesigned sheet is largely effective.
70 % of farmers rated usefulness ≥ 4/5; 67 % could discuss at least one section accurately.

Icons remain the pain-point.
Only 48 % decoded trait icons correctly.

Materials need robustness & production speed.

Diploma (recognition) Matters
Having a tangible certificate made them feel that their effort was officially acknowledged.









Feedback Sessions Conclusions

- The session strengthens trust in both the institutions and the quality of the data.
- The session and feedback sheet provide a summary of the content, which helps reduce the workload for the implementers.
- Implementers need a well-designed and summarized workshop with clear steps, time management, and easy-to-distribute materials.
- IITA and NRCRI teams felt that this time the farmers were heard, and their opinions, not just their data, were taken into account.

Next steps

Design Refinements

- Polish Iconography.
- Review layout and apply minor tweaks.

Local-Language Adaptation

- Find scalable ways to translate sheet to local languages.

ClimMob Integration

- Automate data-pull and one-click generation of individualized sheets.
- Easily extracted workshop structure, formats and materials.

Cross-Context Validation

- Test the package in additional countries and non-cassava crops.
- Compare usability, comprehension, and NPS across contexts.

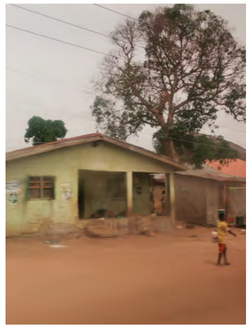
Continuous Research Loop

- Collect feedback metrics each cycle to feed minor updates.

Training

- Run brief virtual/in-person trainings to standardize facilitation steps.







Thank you!

